Mathematical Methods in Uncertainty Quantification (MMUQ) – L06

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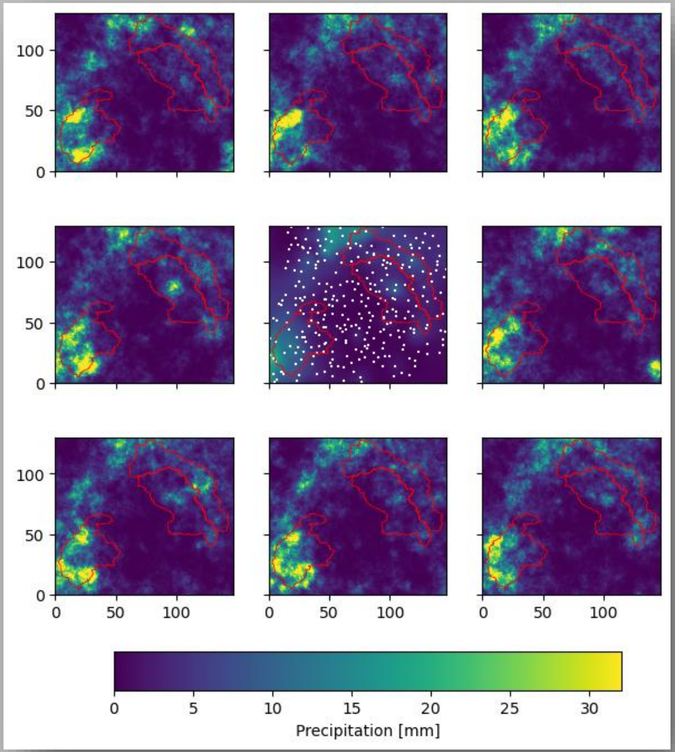
MMUQ L06 – Model Input Uncertainty, and 4th Assignment

- Input Uncertainty
- Assignment
- Tasks

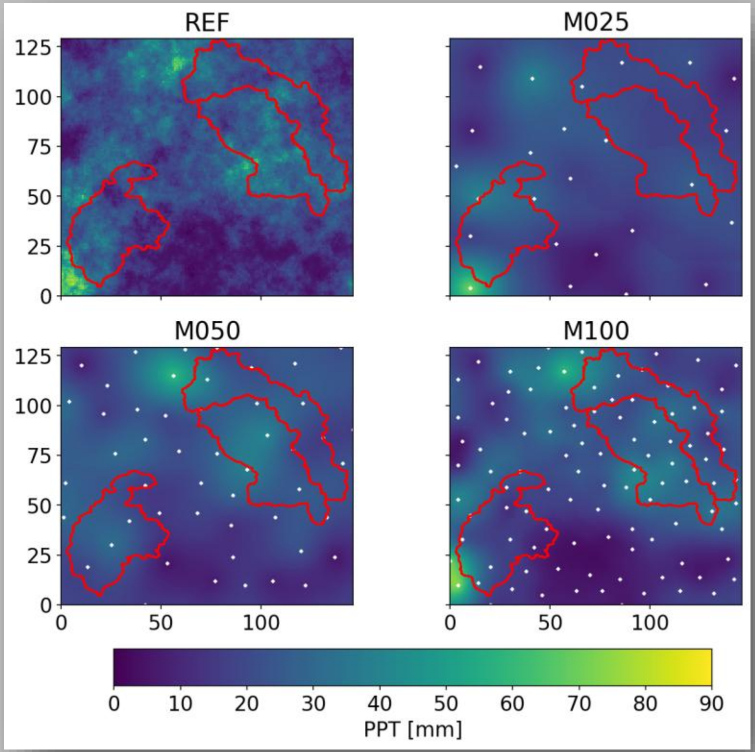
How can be input uncertain?

- Simple measurement errors
- Input continuously distributed in space-time
- Observations are made at points
- What happens at unobserved locations?
  - Fantasy
- Different interpolation schemes
  - Different assumptions
  - Different spatial averages and variances
  - Different relationship to OFV
  - Different calibration parameters

What happens at unobserved locations?



Input uncertainty causes: Varying gauge densities



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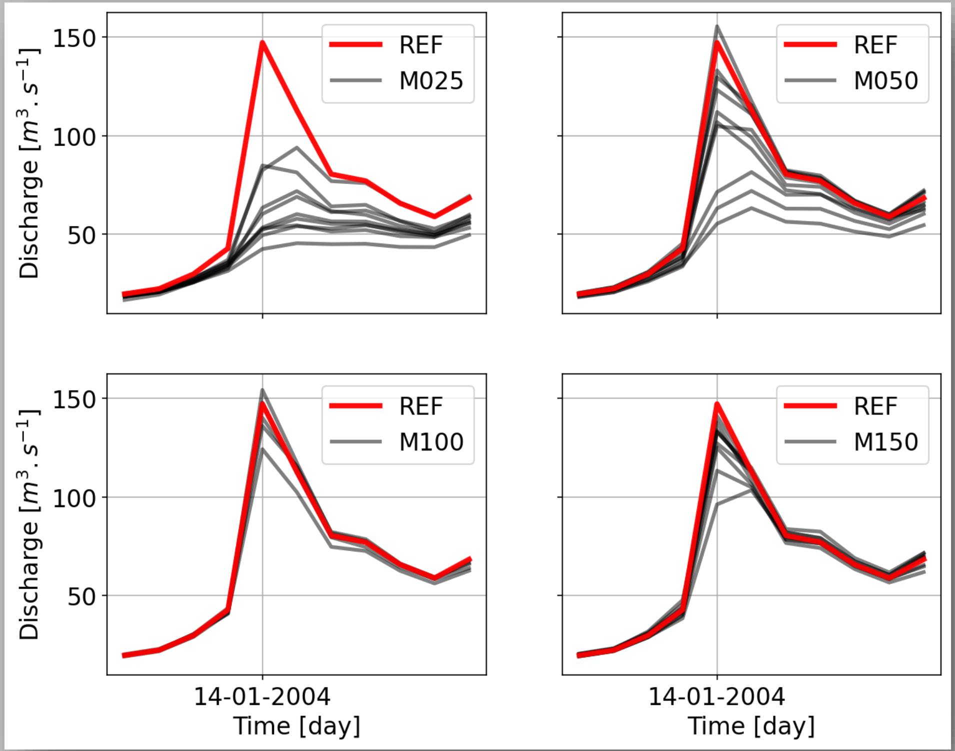

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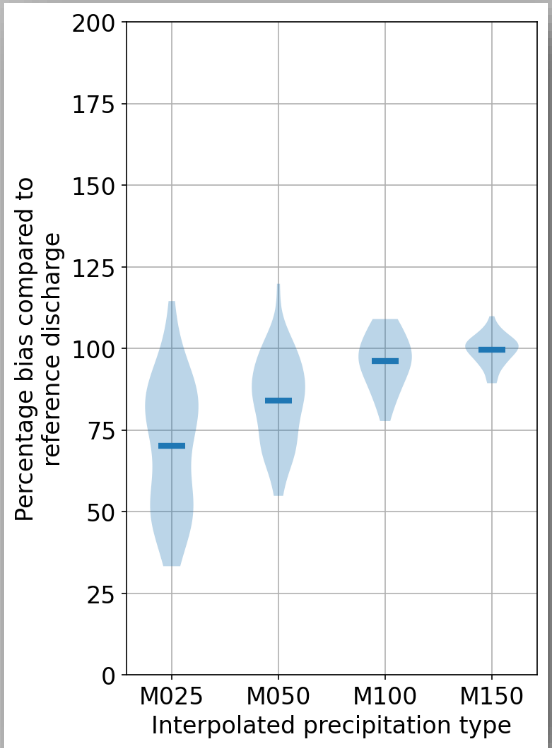
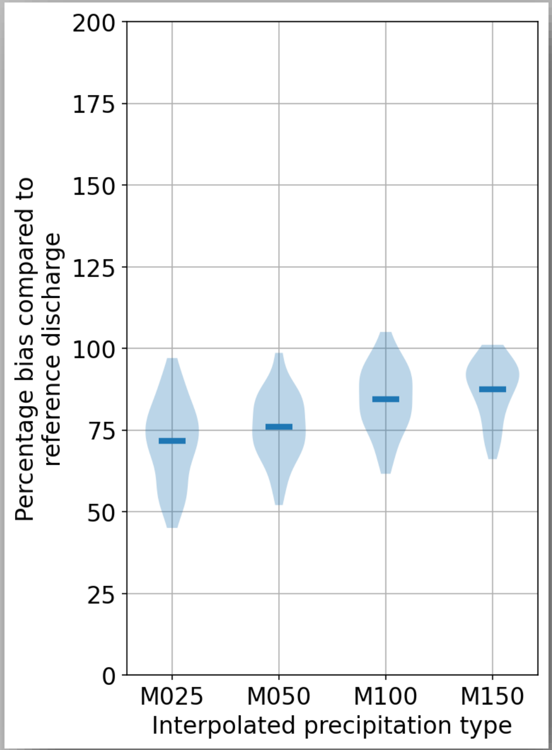


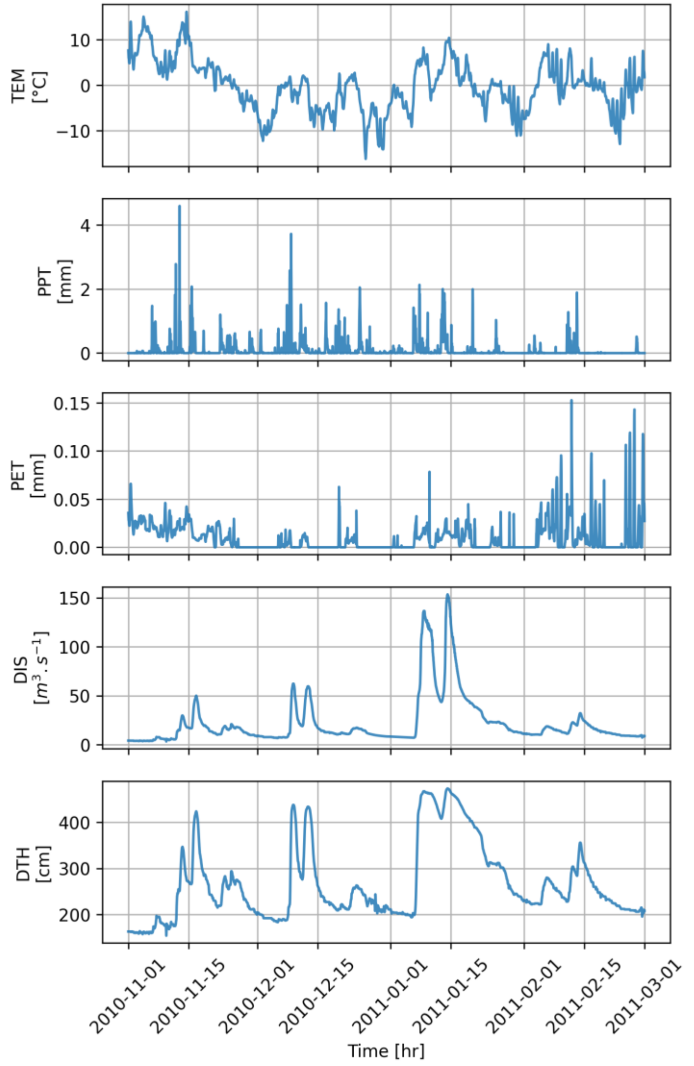


Assignment – 4

- Model output uncertainty due to uncertain inputs
- Perturb inputs systematically
- Analyze effects on model output
- Presentation
  - Make comparisons
    - Reference calibration
    - Recalibrate models
  - Uncertainty bounds of model input
    - Up to 25%
- Reference: Impact of biased and randomly corrupted inputs on the efficiency and the parameters of watershed models, Oudin et al. 2006, Paper

Assignment – 4, The Problem

- Given
  - Reference calibration
  - Original/reference Inputs
- Perturb input values
  - Apply relative random changes
  - Up to 25% (each value)
  - Do 2000 times
- Compare results
  - Use reference parameters
  - Recalibrate (2000 times)



Assignment – 4, Tasks

- Keep it simple
- Create 2000 perturbed series
  - Only PPT
  - Gaussian: C = N(1, 0.05)
  - Scale each value with C
  - Keep all positive!
- Compare cumulative distribution functions (CDFs)
  - Each against the original input
  - Serves as verification
- Run each series with reference parameters
- Recalibrate using each perturbed series
- Compare CDFs
  - Each recalibrated parameter (X)
  - Show reference X
- Compare OFV CDFs
  - Reference and recalibration
  - Reference OFV
- Scatter
  - Relative change against OFV
    - Mean absolute relative PPT change!
  - Using reference and recalibrated Xs
  - Reference OFV

Assignment – 4, Tasks (cont.)

- Scatter OFV
  - Using perturbed inputs
  - Original vs. recalibrated
- Answer
  - How often did the results get better by chance?
    - Compared to reference
    - Both cases
  - How much can recalibration compensate for?
  - Does it make sense to use input uncertainty during calibration?
  - Difference to Oudin et al. 2006?

Group Work in Breakout Rooms

That was it for today, thanks!